

Statistics Teaching System: Architecture and Operations Guide

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1 System Overview

This document describes the complete architecture and operational procedures of the **Statistics Teaching System** (**stat**), designed for the English Taught Program in International Business (ETP) at Shih Chien University. The system supports two sequential courses—*Statistics (1)* and *Statistics (2)*—and is built on a core technology stack of R, Quarto, and R Markdown, integrating website publication, in-class exercises, and automated grading into a unified workflow.

Course website: <https://yyliou.github.io/stat> **Instructor:** Yu-You Liou (d10627008@ntu.edu.tw)

1.1 Course Structure

Table 1: Course Identification and Scheduling

Course	Code	Semester	Schedule	Room	Chapters
Statistics (1)	EIB-21E-01-A1	Fall 2026	Tuesday, Periods 8-9 (15:10-17:00)	A207	Ch. 1-7
Statistics (2)	EIB-21F-01-A1	Spring 2026	Thursday, Periods 6-7 (13:10-15:00)	L309	Ch. 8-12

Each session is structured as two consecutive hours: the first hour is a **Lecture** covering theoretical foundations, and the second hour is a **Lab** in which students apply concepts through in-class exercises using the R programming language.

1.2 Assessment Structure

Table 2: Assessment Weighting and Components

Component	Weight	Details
In-Class Exercises	36%	12 exercises x 3% each
Midterm Examination	32%	
Final Examination	32%	

2 Directory Architecture

2.1 Root Directory Structure

The system root (`stat/`) is organized as follows:

```
stat/
|-- _quarto.yml          # Quarto website global configuration
|-- styles.css           # Custom CSS stylesheet (currently disabled in YAML)
|-- index.html           # Published homepage (rendered output)
|-- sylla.html           # Published syllabus landing page
|-- slide.html           # Published slide index
|-- prob.html            # Published problem set page
|-- r.html               # Published R guide
|-- site_libs/           # Quarto-generated site assets
|
|-- web/                 # Website page sources (qmd)
|   |-- web.Rproj        # RStudio project file
|   |-- index.qmd        # Homepage (course overview, exam regulations)
|   |-- sylla.qmd        # Syllabus landing page (links to PDF syllabi)
|   |-- slide.qmd        # Slide index page
|   |-- prob.qmd         # Problem set page (template, submission links)
|   |-- r.qmd            # R installation and usage guide
|
|-- ch2/ ... ch14/       # Per-chapter lecture slide directories
|   |-- chN.qmd          # English Reveal.js slide source
|   |-- chN-zh.qmd       # Traditional-Chinese parallel version
|   |-- bib.bib          # Chapter bibliography
|   |-- logo.svg         # Slide logo asset
|
|-- stat1-ps/            # Statistics (1) problem sets
|   |-- exN.qmd          # Exercise source (Reveal.js format)
|   |-- exN.html         # Compiled HTML for publication
|
|-- stat2-ps/            # Statistics (2) problem sets (same structure)
|
|-- stat1-ex/            # Statistics (1) student submissions
|   |-- ex1/ ... ex12/
|       |-- exN-[StudentID].pdf
|
|-- stat2-ex/            # Statistics (2) student submissions (same structure)
|
|-- stat1-mid/           # Statistics (1) midterm examination materials
|-- stat1-fin/           # Statistics (1) final examination materials
|-- stat2-mid/           # Statistics (2) midterm examination materials
|-- stat2-fin/           # Statistics (2) final examination materials
|
|-- stu1/                # Statistics (1) roster and grading outputs
|   |-- [timestamp].xls  # Official roster exported from university system
|   |-- ex[N]/
|       |-- ex-score.xlsx
|       |-- stat.Rmd
|       |-- stat.pdf
|
|-- stu2/                # Statistics (2) roster and grading outputs (same structure)
|
|-- sylla/                # Syllabus source files
|   |-- stat1.Rmd / stat1.pdf
|   |-- stat2.Rmd / stat2.pdf
```

```
|
|-- !-Archive/          # Archive of retired materials
|
`-- prompt/             # AI-assisted prompt documents
    |-- slide-format.Rmd # Master specification for slide generation
    |-- prompt.Rmd       # Legacy slide generation prompt
    |-- ex-review1.Rmd   # Grading prompt for Statistics (1)
    |-- ex-review2.Rmd   # Grading prompt for Statistics (2)
    |-- render.txt       # Batch render command for all chapter slides
    |-- bluman-step-by-step-statistics-8th-edition.pdf # Textbook
    `-- stat-system-guide.Rmd # This document
```

2.2 Design Principles

The directory structure adheres to six organizational principles:

1. **One directory per chapter.** Directories `ch2/` through `ch14/` are self-contained units (sources, bibliography, logo, rendered output), facilitating independent version control and selective rendering.
2. **Bilingual slide parity.** Every chapter maintains an English source (`chN.qmd`) and a structurally identical Traditional-Chinese version (`chN-zh.qmd`), cross-linked from the title slide.
3. **Separation of page sources and published output.** Website page sources live in `web/`; rendered HTML is published from the repository root for GitHub Pages.
4. **Separation of problem sets and submissions.** `stat1-ps/` (instructor-authored questions) and `stat1-ex/` (student submissions) maintain distinct responsibilities and access patterns.
5. **Co-location of rosters and grading outputs.** Each `stu1/` or `stu2/` directory holds both the official class roster and all automated grading artifacts, enabling single-location lookup.
6. **Centralized prompt management.** All AI-assisted documents, including the textbook PDF used as the generation source, are isolated within `prompt/`, decoupled from pedagogical content.

3 Technology Stack

3.1 Core Tools

Table 3: Core Software Components

Tool	Minimum Version	Purpose
R	≥ 4.2	Statistical computation and document generation
RStudio Desktop	Latest stable	Integrated development environment
Quarto	≥ 1.3	Website and slide publication
R Markdown	≥ 2.20	Assignment templates and grading reports
TinyTeX	Current	LaTeX engine for PDF output

3.2 Required R Packages

The following packages must be installed prior to system operation:

```
# Document generation
install.packages(c("rmarkdown", "knitr", "tinytex"))
tinytex::install_tinytex()

# Data manipulation and visualization
install.packages(c("tidyverse", "ggplot2", "patchwork"))

# Spreadsheet I/O for grading reports
install.packages(c("readxl", "openxlsx"))

# Academic integrity (UUID generation)
install.packages("uuid")
```

3.3 Website Publication Architecture

The course website is built as a **Quarto Website** project, governed by `_quarto.yml`. Key configuration decisions are summarized below.

Table 4: Quarto Website Configuration Summary

Setting	Value
output-dir	. (HTML output at repository root for GitHub Pages)
Page sources	web/ (index, sylla, slide, prob, r)
Theme	Minty + brand (styles.css currently disabled)
Navigation bar	Syllabus -> Slides -> Problem sets -> R
Slide format	Reveal.js
Math rendering	KaTeX

4 Lecture Slide Authoring Workflow

4.1 File Format

Each chapter's slides reside in `chN/chN.qmd` (English) and `chN/chN-zh.qmd` (Traditional Chinese). The YAML header is **fixed** by the master specification (`prompt/slide-format.Rmd`); only the subtitle and the language-switch link target may vary:

```
---
title: "Statistics"
subtitle: "Chapter N: [Title]"
author: "Yu-You Liou"
institute: "Shih Chien University"
date: "2026-06-07"
format:
  revealjs:
    html-math-method: katex
    smaller: true
    theme: serif
    footer: "Elementary Statistics: A Step by Step Approach"
    logo: "logo.svg"
    editor: source
    slide-number: true
    scrollable: false
    transition: "none"
    menu:
      useTextContentForMissingTitles: false
      hideMissingTitles: true
      side: 'left'
      titleSelector: 'h1'
---
```

The header additionally includes an `include-after-body` script and title-slide attributes that inject a language-switch link (`chN-zh.html` in the English file; `chN.html` in the Chinese file).

4.2 Authoring Standards

All slide content must conform to the master specification in `prompt/slide-format.Rmd`, which is the single source of truth. Key conventions:

- Each slide begins with a level-2 heading (`##`).
- Mathematical expressions use standard LaTeX syntax, e.g., $\bar{x} = \frac{\sum x}{n}$.
- Definitions and key formulae are enclosed in Quarto callout blocks, which **must be perfectly balanced** (unbalanced callouts are the most common rendering failure):

```
::: {.callout-note}
**Definition** ...
:::
```

- Every textbook example provides two parallel solutions: (1) a complete algebraic derivation in LaTeX, and (2) an executable R implementation with displayed output.
- All prose must be original; verbatim reproduction of textbook text is prohibited.
- **Bilingual parity:** `chN.qmd` and `chN-zh.qmd` must be structurally identical, slide for slide.
- Chapters `ch2/–ch5/` serve as the clean reference standard for layout and style.

4.3 Generating a New Chapter

Step 1. Create the chapter directory with shared assets:

```
mkdir chN/ and copy bib.bib and logo.svg from an existing chapter.
```

Step 2. Provide `prompt/slide-format.Rmd` to an AI assistant and request `chN.qmd` and `chN-zh.qmd` for the target textbook chapter

(textbook PDF available in prompt/). The agent must complete the validation checklist in the specification before delivery.

Step 3. Render both language versions (see prompt/render.txt for the batch command):

```
quarto render chN/chN.qmd
```

```
quarto render chN/chN-zh.qmd
```

Step 4. Update the chapter index in web/slide.qmd.

5 Problem Set Management

5.1 Directory Mapping

Table 5: Problem Set and Submission Directory Mapping

Course	Problem Set Directory	Submission Directory
Statistics (1)	stat1-ps/	stat1-ex/
Statistics (2)	stat2-ps/	stat2-ex/

5.2 Problem Set Format

Problem sets are authored as Reveal.js presentations (`stat1-ps/exN.qmd`), with one question per slide. A representative header:

```
---
title: "Statistics 1"
subtitle: "In-Class Exercise N"
author: "Yu-You Liou"
date: today
institute: "Shih Chien University"
format:
  revealjs:
    html-math-method: katex
---
```

5.3 Submission and Grade Distribution Channels

Submission and grade lookup are handled through shared **pCloud** links published on the problem set page (`web/prob.qmd`): one upload link per course (students place the PDF in the matching `exN` folder, which syncs to `stat[N]-ex/exN/`), and one read-only link per course for checking grades.

5.4 Submission Requirements

Students must use the official template below and upload to the course pCloud folder `exN/` (synced to `stat1-ex/exN/`) using the exact naming convention `exN-[StudentID].pdf`.

```
---
title: "Ex. N"
author: "StudentID"
date: "2026-06-07"
output: pdf_document
---
```

```
\` \` \`{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
\` \` \`

\` \` \`{r}
print(uuid::UUIDgenerate())
\` \` \`
```

Every submission must include the UUID generation code block. This serves as the primary mechanism for academic integrity verification.

6 Automated Grading System

6.1 Process Overview

The grading pipeline consists of four sequential stages:

```

STAGE 1 | Initialization and Roster Reconciliation
  Read stu[N]/[timestamp].xls -> construct student roster
  Cross-reference stat[N]-ex/exN/ -> flag absent students
  as Missing (Score = 0)

STAGE 2 | Compliance Filtering (sequential; stops at first violation)
  (i) Submission timestamp > [deadline] (UTC+8)
      -> Late Submission (Score = 0)
  (ii) File is not PDF or does not match exN-[StudentID].pdf
      -> Incorrect File Name/Format (Score = 0)
  (iii) YAML missing title/author/date, or UUID chunk absent,
        or author field does not match submitting student ID
      -> Incorrect Internal Format (Score = 0)
  (iv) UUID value identical to another submission, or
        invalid/forged UUID
      -> Plagiarism (Score = 0; all implicated students)

STAGE 3 | Academic Content Grading
  Total: 100 points, allocated equally across N questions
  Each question scored against stat[N]-ps/exN.qmd answer key
  Incorrect items noted as: [QN] is incorrect

STAGE 4 | Output Generation
  stu[N]/exN/ex-score.xlsx <- Scoreboard (4 columns)
  stu[N]/exN/stat.Rmd      <- Analytics report source
  stu[N]/exN/stat.pdf      <- Compiled analytics report

```

6.2 Scoreboard Format (ex-score.xlsx)

Table 6: Scoreboard Column Specification

Column	Type	Description
StudentID	Character	University student identification number
Name	Character	Student full name as listed in official roster
Score	Integer (0-100)	Final score awarded for the submission
Note	Character	Compliance violation code or incorrect question numbers

6.3 Analytics Report (stat.Rmd)

The analytics report reads `ex-score.xlsx` and produces three outputs:

1. **Descriptive statistics table:** minimum, Q_1 , median, Q_3 , maximum, and mean of the score distribution.
2. **Boxplot:** a publication-quality visualization of the score distribution rendered via `ggplot2`.
3. **Submission summary table:** counts for Total, Missing, Late Submission, Incorrect Format, and Plagiarism.

6.4 Grading Note Codes

Table 7: Grading Note Code Reference

Code	Description	Score
Missing	No file submitted by the deadline	0
Late Submission	File submitted after the official deadline	0
Incorrect File Name/Format	File is not PDF or does not follow naming convention exN-[ID].pdf	0
Incorrect Internal Format	YAML header incomplete, author ID mismatch, or UUID chunk absent	0
Plagiarism	UUID value matches another submission, or is invalid/forged	0
[QN] is incorrect	Question N answer is incorrect or methodology is flawed	Partial deduction

7 Academic Integrity Mechanisms

7.1 UUID Verification

Every submission must execute and display output from the following R code block:

```
print(uuid::UUIDgenerate())
```

`uuid::UUIDgenerate()` produces a Version 4 UUID (Universally Unique Identifier) on each invocation, e.g., `6d2b9fda-2d9b-48f3-84bc-f82eabc39b4d`. The grading system compares this value across all submissions for a given exercise. If two or more files share an identical UUID, or if a file's UUID is invalid or forged, all implicated students receive a score of zero with the note **Plagiarism**.

Important implementation notes:

- The compliance filter is applied sequentially per file. A file that fails Stage 2 check (iii) (Incorrect Internal Format) is not evaluated for plagiarism within its own record; however, its UUID remains in the comparison pool and may trigger a Plagiarism flag in other students' records.
- The `author` field in the YAML header must contain the submitting student's own ID. Use of another student's ID constitutes a format violation independent of the plagiarism check.

7.2 Examination Regulations

In alignment with the course's paperless policy, examinations are conducted online (a sample examination interface is linked from the course homepage). Examination materials are maintained in `stat1-mid/`, `stat1-fin/`, `stat2-mid/`, and `stat2-fin/`.

Table 8: Examination Conduct Regulations

Regulation	Policy
Mandatory attendance	Roll call 5 minutes before start; late arrival or absence results in score = 0
Network restriction	System tracks IP addresses; campus Wi-Fi required; non-campus IP results in score = 0
Authorized resources	Open-book; reference materials and AI tools are permitted as auxiliary aids
Media prohibition	Photography, video/audio recording, and screenshotting result in score = 0
Communication prohibition	Interpersonal communication and disruptive behavior result in score = 0
Copy-paste disabled	Copy-paste functionality is disabled by the examination system

7.3 AI Use Policy

Students may use AI tools (e.g., ChatGPT, Gemini) as learning aids. However, students must be capable of explaining and defending any code or analysis they submit. Uncritical submission of AI-generated content without demonstrated understanding constitutes academic dishonesty and is subject to the same consequences as plagiarism.

8 AI Prompt Management

The `prompt/` directory contains structured prompt documents for AI-assisted course material generation and grading.

Table 9: Prompt Directory File Inventory

File	Purpose
<code>slide-format.Rmd</code>	Master specification for AI-assisted slide generation (single source of truth; supersedes <code>prompt.Rmd</code>)
<code>prompt.Rmd</code>	Legacy slide generation prompt (retained for reference)
<code>ex-review1.Rmd</code>	Automated grading prompt for Statistics (1) exercises
<code>ex-review2.Rmd</code>	Automated grading prompt for Statistics (2) exercises
<code>render.txt</code>	Batch quarto render command for all chapter slides
<code>bluman-...-8th-edition.pdf</code>	Source textbook PDF used for slide generation
<code>stat-system-guide.Rmd</code>	This technical document

8.1 Slide Generation Specification (`slide-format.Rmd`)

`slide-format.Rmd` is a complete, self-contained specification pasted to the AI agent together with the instruction “Generate *chX.qmd* (and *chX-zh.qmd*) for Chapter X following this specification exactly.” Key directives:

- **Fixed YAML front matter:** Only the subtitle and the language-switch link may vary; everything else is reproduced verbatim.
- **Structural correctness first:** Decks must render without error; every callout block must be perfectly balanced (the most common failure mode).
- **Exhaustive extraction:** Every example in the target chapter must be incorporated; summarization or omission is not permitted.
- **Dual-solution requirement:** Each example must present (1) a full algebraic derivation in LaTeX and (2) executable R code with displayed output.
- **Paraphrasing mandate:** All prose must be rewritten in original language; verbatim textbook reproduction is prohibited.
- **Bilingual parity:** Each chapter ships as a structurally identical English/Traditional-Chinese pair.
- **Reference standard:** Chapters `ch2–ch5` are the clean style reference.
- **Validation checklist:** The agent must run the specification’s built-in checklist before returning a file.

8.2 Grading Prompts (`ex-review1.Rmd`, `ex-review2.Rmd`)

Key components of the grading prompts:

- Definition of environment variables: `[deadline]` (UTC+8), `[no]` (exercise number), and root directory structure.
- Specification of all four grading stages with precise conditional logic.
- Output format requirements: scoreboard column names, Rmd content sections, and PDF compilation.

The two files are identical except for the directory targets (`stu1/stat1-ps/stat1-ex` vs. `stu2/stat2-ps/stat2-ex`).

9 Operations Guide

9.1 New Semester Initialization

1. Export official class roster from the university administrative system. Save to `stu1/` (Statistics 1) or `stu2/` (Statistics 2).
2. Update `sylla/stat1.Rmd` and `sylla/stat2.Rmd` with the current semester's schedule and policies. Knit to PDF.
3. Verify `_quarto.yml` navigation links and semester metadata in `web/index.qmd` and `web/sylla.qmd`.
4. Create or empty the pCloud submission folders (`ex1-ex12`) and confirm the upload/grade-lookup links in `web/prob.qmd`.
5. Rebuild the website pages and republish.

9.2 Weekly Problem Set Publication

1. Create `stat1-ps/exN.qmd` (or `stat2-ps/exN.qmd`) in Reveal.js format. Each slide corresponds to one question.
2. Render the problem set:
`quarto render stat1-ps/exN.qmd`
3. Update the problem set index in `web/prob.qmd`.

9.3 Weekly Submission Collection and Grading

1. Students upload `exN-[StudentID].pdf` to the course pCloud folder, which syncs to `stat1-ex/exN/`.
2. Execute the AI grading prompt (`prompt/ex-review1.Rmd`) with `[no]` set to the exercise number and `[deadline]` set to the submission deadline (UTC+8).
3. Verify generated scoreboard: `stu1/exN/ex-score.xlsx`
4. Verify analytics report: `stu1/exN/stat.pdf`
5. Grades are published to students via the read-only pCloud link.

9.4 Generating a New Lecture Chapter

1. Create `chN/` and copy `bib.bib` and `logo.svg` from an existing chapter.
2. Generate `chN.qmd` and `chN-zh.qmd` using `prompt/slide-format.Rmd` with an AI assistant, referencing the textbook PDF in `prompt/`.
3. Review, edit, and finalize both files in RStudio.
4. Render (batch command in `prompt/render.txt`):
`quarto render 'ch*/ch*.qmd'`
5. Add the chapter link to `web/slide.qmd`.

9.5 Full-Site Publication (GitHub Pages)

```
# Render updated pages/slides, e.g.
quarto render 'ch*/ch*.qmd'

# Commit and push to trigger GitHub Pages deployment
git add .
git commit -m "Update: Week N - [brief description]"
git push origin main
```

9.6 Troubleshooting Reference

Table 10: Common Issues and Resolutions

Symptom	Probable Cause	Resolution
PDF output fails	TinyTeX not installed or incomplete	Run: <code>tinytex::install_tinytex()</code>
UUID package not found	uuid package not installed	Run: <code>install.packages('uuid')</code>
Quarto render error	Quarto version outdated	Update Quarto to latest stable release
Roster file read error	Legacy .xls format compatibility	Use readxl package or resave as .xlsx
Slide math not rendering	KaTeX configuration missing	Set <code>html-math-method: katex</code> in YAML

10 Course Chapter Reference

10.1 Statistics (1): Chapter Schedule

Table 11: Statistics (1) Weekly Schedule and Exercise Mapping

Week	Chapter	Topic	Exercise
1	—	Syllabus and Class Regulations	—
2	Ch. 1	The Nature of Probability and Statistics	ex1
3	Ch. 2	Frequency Distributions and Graphs	ex2
4	Ch. 3	Data Description	ex3
5	Ch. 3	Data Description (continued)	ex4
6	Ch. 4	Probability and Counting Rules	ex5
7	Ch. 4	Probability and Counting Rules (continued)	ex6
8	—	Midterm Review	—
9	—	Midterm Examination	—
10	Ch. 5	Discrete Probability Distributions	ex7
11	Ch. 5	Discrete Probability Distributions (continued)	ex8
12	Ch. 6	The Normal Distribution	ex9
13	Ch. 6	The Normal Distribution (continued)	ex10
14	Ch. 7	Confidence Intervals and Sample Size	ex11
15	Ch. 7	Confidence Intervals and Sample Size (continued)	ex12
16	—	Final Examination	—
17-18	—	Flexible Learning Weeks	—

10.2 Statistics (2): Chapter Schedule

Table 12: Statistics (2) Weekly Schedule and Exercise Mapping

Week	Chapter	Topic	Exercises
1	—	Syllabus and Class Regulations	—
2-4	Ch. 8	Hypothesis Testing	ex1-3
5-6	Ch. 9	Testing the Difference Between Two Means, Two Proportions, and Two Variances	ex4-5
7	Ch. 10	Correlation and Regression	ex6
8	—	Midterm Review	—
9	—	Midterm Examination	—
10-11	Ch. 10	Correlation and Regression (continued)	ex7-8
12-13	Ch. 11	Other Chi-Square Tests	ex9-10
14-15	Ch. 12	Analysis of Variance	ex11-12
16	—	Final Examination	—
17-18	—	Flexible Learning Weeks	—

Note: Slide decks for Chapter 13 (Nonparametric Statistics) and Chapter 14 (Sampling and Simulation) are maintained in the repository and listed on the slide index as supplementary material, but are not part of the current Statistics (2) schedule.

11 Appendix

11.1 A. Official Submission Template

```

---
title: "Ex. 1"
author: "A113290001"
date: "2026-09-22"
output: pdf_document
---

# This chunk is mandatory in every submission.
print(uuid::UUIDgenerate())

# Exercise solutions follow below.

```

11.2 B. Directory Quick Reference

Table 13: Directory Purpose Quick Reference

Directory	Purpose
web/	Website page sources (index, sylla, slide, prob, r) and RStudio project
ch2/-ch14/	Bilingual Reveal.js lecture slide sources, one subdirectory per chapter
stat1-ps/	Statistics (1) in-class exercise problem sets
stat2-ps/	Statistics (2) in-class exercise problem sets
stat1-ex/exN/	Statistics (1) student PDF submissions for exercise N
stat2-ex/exN/	Statistics (2) student PDF submissions for exercise N
stat1-mid/, stat1-fin/	Statistics (1) midterm/final examination materials
stat2-mid/, stat2-fin/	Statistics (2) midterm/final examination materials
stu1/exN/	Statistics (1) grading outputs (scoreboard, analytics report)
stu2/exN/	Statistics (2) grading outputs (scoreboard, analytics report)
sylla/	Syllabus R Markdown source files
prompt/	AI prompts, master slide specification, textbook PDF
!-Archive/	Archive of retired materials

11.3 C. References

- Bluman, A. G. (2012). *Elementary Statistics: A Step-by-Step Approach* (8th ed.). McGraw-Hill.
- R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org>
- Posit, PBC. (2024). *RStudio Desktop*. <https://posit.co/download/rstudio-desktop>
- Allaire, J. J., et al. (2024). *Quarto*. <https://quarto.org>
- Course website: <https://yyliou.github.io/stat>